◦ Defn.

◦ A complex number is of the form $x+iy$ where $x, y \in \mathbb{R}$.

$z = x+iy$ $\quad x =$ "real part of z" $= Re(z)$

$\quad\quad\quad y =$ "imaginary part of z" $= Im(z)$

* $\mathbb{C} \to \mathbb{R}^2$

$z \mapsto (x,y)$

$i \mapsto (0,1)$



"The Complex Plane"

* If $Im\, z = 0$, z is real

$Re\, z = 0$, z is pure imaginary

* Operations on $\mathbb{C}$: let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$

I.) $z_1 + z_2 = (x_1+x_2) + i(y_1+y_2)$

II.) $z_1 \cdot z_2 = (x_1 \cdot x_2 - y_1 y_2) + i\,(x_1 y_2 + x_2 \cdot y_1)$

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* NOTE !

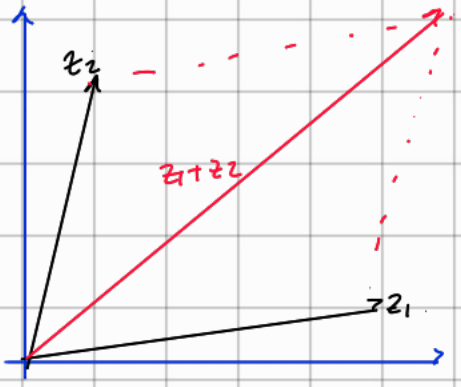
* If $i^2 = -1$, $z_1 \cdot z_2$ is usual multiplication

$(x_1 + iy_1)\cdot(x_2 + iy_2) = x_1x_2 + x_1 iy_2 + x_2 iy_1 + i^2 y_1 y_2$

$= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$

$\mathbb{C} \longleftrightarrow \mathbb{R}^2$

$z_1 + z_2 \longleftrightarrow (x_1+x_2, y_1+y_2)$



* $(\mathbb{C}, +, \cdot)$ is a field.

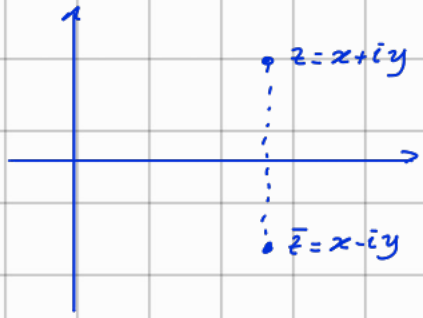
◦ $-z$ is the additive inverse

◦ $1/z$ is the multiplicative inverse

◦ Triangle inequality: $||z|-|w|| \le |z+w| \le |z|+|w|$

◦ $\bar{z} \equiv$ "The conjugate of $\bar{z}$" $= x - iy$

$|z| = \sqrt{x^2+y^2}$ where $z = x+iy$

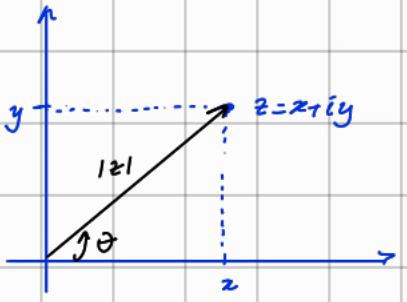


+ Identities / Properties +

+ $z^{-1} = \frac{1}{z} = \frac{1}{x+iy} \cdot \frac{(x-iy)}{(x-iy)} = \frac{x-iy}{x^2+y^2} = \left(\frac{x}{x^2+y^2}\right) + i\left(\frac{-y}{x^2+y^2}\right) = \frac{\bar{z}}{z\,\bar{z}} = \frac{\bar{z}}{|z|^2}$

" $\oint_C \frac{x}{x^2+y^2}\,dy - \frac{y}{x^2+y^2}\,dx$ "

+ $\overline{z+w} = \bar{z} + \bar{w}$
+ $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$
+ $|z| = |\bar{z}|$
+ $\mathrm{Re}\, z = \frac{z+\bar{z}}{2}$ $\quad \mathrm{Im}\, z = \frac{z-\bar{z}}{2i}$
+ $|z-w|$ = the distance between z & w
+ $|z_1+z_2+\cdots+z_n| \le |z_1|+|z_2|+\cdots+|z_n|$



$z = |z|\cdot\cos\theta + i\cdot|z|\sin\theta$

Say $|z| = r$

$\Rightarrow z = r(\cos\theta + i\sin\theta)$

" Euler's formula "

$$e^{i\theta} = \cos\theta + i\sin\theta$$

proof ↓

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n}}{2n!} + i\sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!}$$

$\sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n}}{2n!} = \cos\theta$ $\quad i\sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} = \sin\theta$

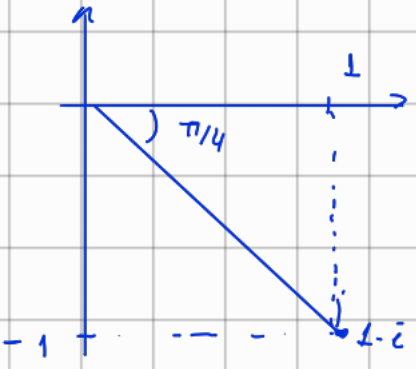
$\Rightarrow z = r\cdot e^{i\theta}$ $\quad |z| = r$, $\quad \theta =$ " the argument of z "

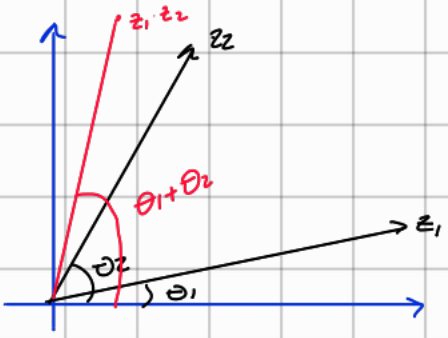
* $-\pi < \theta \le \pi$ $\quad \theta = \mathrm{Arg}\, z$ = " the principle argument "

+ $\arg z = \mathrm{Arg}\, z + 2n\pi$; $n \in \mathbb{Z}$.

Ex.

$z = 1 - i = \sqrt{2}\cdot e^{i(-\pi/4)}$





$z_1 = r_1 \cdot e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$ $\quad z_1 z_2 = (r_1 r_2) e^{i(\theta_1+\theta_2)}$

- $e^{i2\theta} = \cos(2\theta) + i\sin(2\theta) = (e^{i\theta})^2$ $\quad \Rightarrow \arg(z_1 z_2) = \arg z_1 + \arg z_2 + 2n\pi$

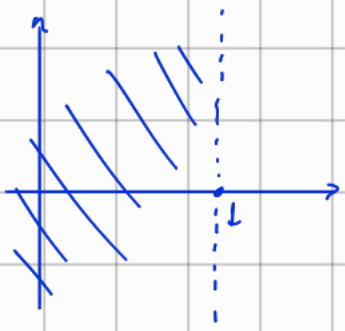
  $= (\cos\theta + \sin\theta)^2 = \cos^2\theta - \sin^2\theta + i(2\cos\theta\sin\theta)$

- $z^n = r^n e^{in\theta} \Rightarrow |z^n| = r^n$

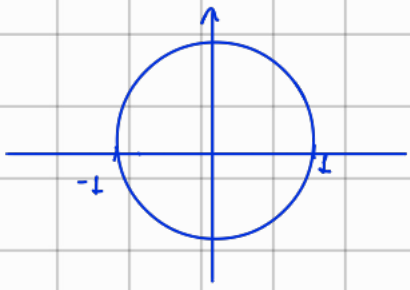
Ex:

Represent the following sets in $\mathbb{C}$

- $\{z / \operatorname{Re} z < 1\}$



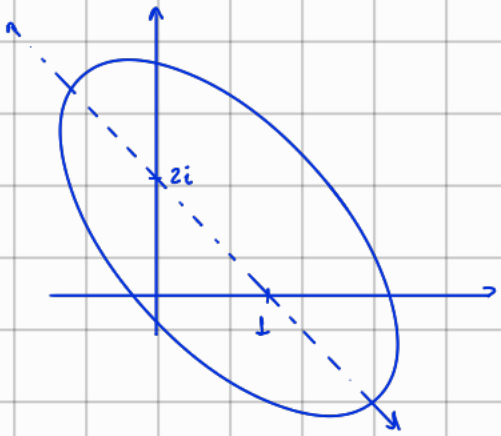
- $\{z / |z| = 1\}$



- $\{z / |z-2i| < 1\}$



- $\{z / |z-1| + |z-2i| = 5\}$



Roots of Complex Numbers